Parts to buy

ModalAI Starling 2 Max — LED strip, cooling fan, propellers. Each item has its own page with a picture, exact specs and a box to write prices in.
Keep every receipt — ask for an itemised invoice showing the shop's name and GST registration number.

✓	Item	Key spec	Where
☐	**Addressable LED strip** **幻彩灯带**	**WS2812B or SK6812** 5 V · 30 LED/m · need ~24 pixels (≈80 cm)	Sim Lim **Tower**
☐	**Cooling fan** **散热风扇**	**25 × 25 × 7 mm · 5 V · 2 WIRES ONLY** ≤ 150 mA	Sim Lim **Tower**
☐	**JST SH 1.0 mm connector** **JST SH 1.0mm 2P 插头线**	2-pin, with pigtail wires (×2–3 spare)	Sim Lim **Tower**
☐	**Resistors** **电阻**	330 Ω or 470 Ω, 1/4 W (×10)	Sim Lim **Tower**
☐	**Hook-up wire** **细电线**	Thin stranded, 3 colours, ~2 m each	Sim Lim **Tower**
☐	**Propellers** **螺旋桨 三叶 7寸**	**7 inch · 3.7 pitch · 3 blades** (7037-3 / 7×3.7×3)	Sim Lim **Square**
☐	*Optional:* 5 V BEC / regulator **5V 稳压模块**	5 V out, ≥ 1 A — only if LED rail measures 3.3 V	Sim Lim **Tower**

Two buildings, side by side

Sim Lim TOWER

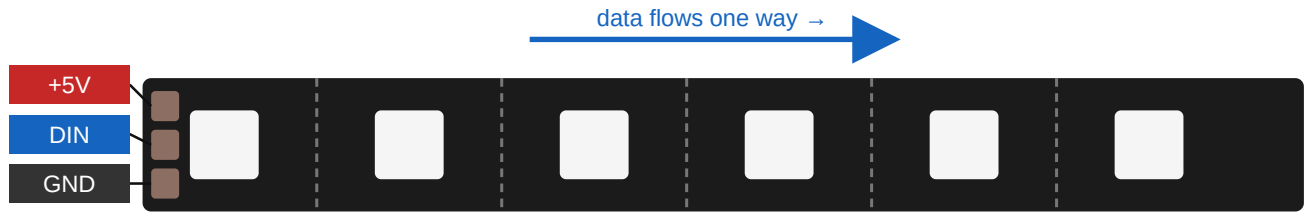
10 Jalan Besar
Electronic components
LED strip · fan · connectors · resistors · wire
Closes ~7:30 pm.

Sim Lim SQUARE

1 Rochor Canal Road
Drone / FPV parts
Propellers — ask for **Hobby Square**
Mon–Sat 12:30–7:30 pm.

1 · LED strip

幻彩灯带 / 可编程灯带



3 solder pads

cut on the dashed lines · 30 LED per metre ≈ 33 mm spacing



This is what to ask for — a flexible addressable strip

Photo: Cytron Technologies, product DS-6812-WS-601M



Count the wires: THREE. Typically red = +5 V, green = data, white = GND.

Photo: Cytron Technologies

WS2812B or SK6812

5 V · **30 LED / metre** · need **24 pixels ≈ 80 cm** (1 metre is fine)

THESE ARE FINE ✓	
Chipset	**WS2812B** / **SK6812** (RGB or RGBW both OK)
Brand	Any
PCB / jacket	Any colour, any IP rating
Reel length	Any — 1 m is plenty

DO NOT BUY ✗	
APA102 / DotStar	needs a clock wire
SK9822	needs a clock wire
WS2801	needs a clock wire
WS2815	12 V — wrong voltage
Non-addressable RGB	cannot address pixels

The test question to ask the shop:

"Is this **individually addressable**, **5 volt**, with **3 wires** — data, +5 V, ground?"

"这个是单点控制的吗？5V的？三线：数据、正极、负极？"

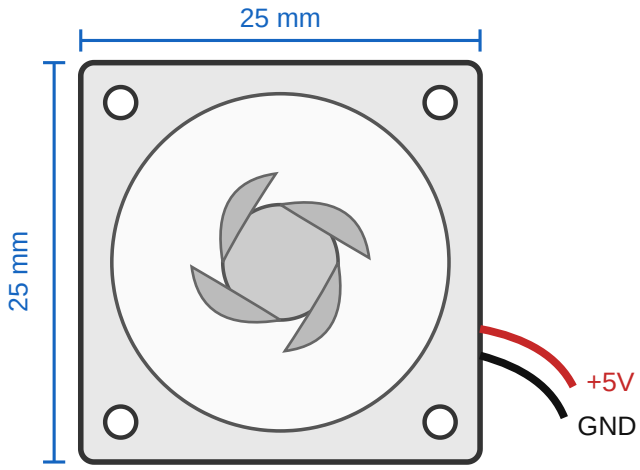
If it needs **4 wires** (extra clock line) it is the wrong type.

Quotes — write the price here

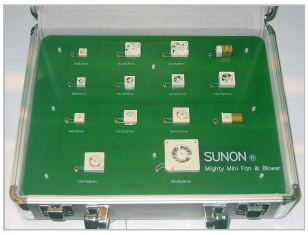
Shop / unit no.	Exact item seen	Price S$	Bought?

2 · Cooling fan

散热风扇 25mm 5V



25 × 25 mm, 7 mm thick, exactly two wires



Sunon mini-fan range — the family ModalAI specifies. Yours is **25 × 25 × 7 mm**; note every one has **2 wires**.

Photo: Rico Shen, CC BY-SA 4.0, via Wikimedia Commons

> **25 × 25 × 7 mm**
> **5 V DC** · **2 WIRES ONLY** · ≤ 150 mA
> Size code: **“2507”**

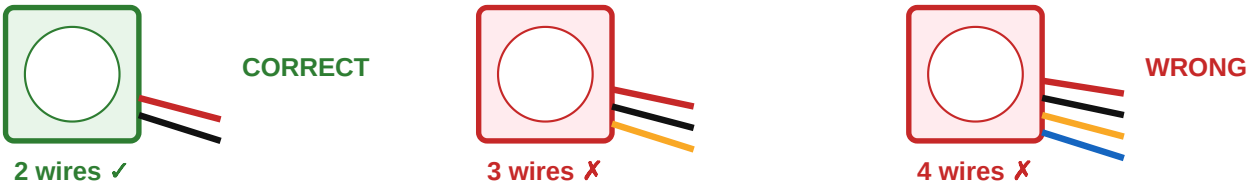
*Shops know this size as **2507**. Asking for “a 2507 5 volt 2-wire fan” is the fastest way to be understood.*

DO NOT BUY ✗	
12 V fan	wrong voltage
3-wire (has tacho)	will not work
4-wire (PWM input)	will not work

*Prefer a **ball bearing** fan (滚珠轴承) over sleeve — it is going on a flying drone and will be vibrating.*

*30 mm or 20 mm is acceptable **only** if 25 mm is unavailable — ask me before buying a different size.*

Why the wire count matters



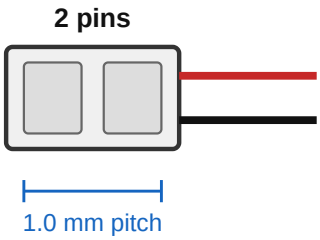
*The drone switches the fan's **ground** on and off to control speed. A 3- or 4-wire fan has extra pins that need a constant ground, so it will stutter or not start.*

Quotes — write the price here

Shop / unit no.	Exact item seen	Price S$	Bought?

3 · Connector, resistors, wire

插头 · 电阻 · 电线





orange-orange-brown = 330 Ω



This shape — 2-pin housing with two wires already crimped on

Photo: Christopher.nunez, CC BY-SA 3.0, via Wikimedia Commons. Shown for shape only — confirm the pitch is 1.0 mm.

JST SH 1.0 mm

2-pin, female, **with wires attached**
(a “pigtail” / 带线插头)

*This is the **small** JST — 1.0 mm between pins. Do **not** buy JST PH (2.0 mm) or XH (2.5 mm); they are bigger and will not fit.*
Buy 2–3 spares, they are easy to damage.

330 Ω or 470 Ω

1/4 watt, ordinary carbon film
Buy **10** — they cost cents

Either value works. Anything between 300 and 500 Ω is fine.

Hook-up wire

Thin stranded, **3 colours** (red / black / any)
~2 m of each · 24–28 AWG

Optional — only if needed

5 V BEC / regulator module

5 V output, 1 A or more. Input must accept **7–17 V** (it runs from a 4S battery).

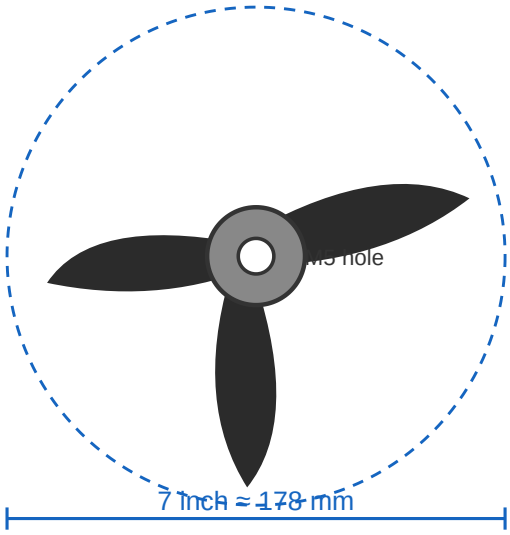
Buy this only if the drone's LED power pad measures 3.3 V instead of 5 V. If unsure, skip it — it can be bought later.

Quotes — write the price here

Shop / unit no.	Exact item seen	Price S$	Bought?

4 · Propellers

螺旋桨 · 三叶 · 7寸



Three blades. Count them.



The exact bag to look for. Label reads **GEMFAN 7037-3**, “Black Reinforced Carbon Nylon (2L+2R)”.

Photo: OddityRC / Gemfan product listing

7 × 3.7 × 3

7 inch diameter (≈178 mm)
3.7 inch pitch
3 blades (tri-blade)

Ask for	**“7037” or “7×3.7 tri-blade”** e.g. Gemfan 7037-3
Bore	M5 (5 mm centre hole)
Quantity	**2 sets of 4** (CW and CCW)
Material	Nylon / carbon-nylon — either

PITCH MUST NOT BE HIGHER THAN 3.7

Do not accept 7040, 7042, 7045 or any “freestyle” prop. A higher-pitch prop will overheat the motors and ESC on this aircraft.

螺距不可以超过 3.7

7035 (3.5 pitch) is acceptable as a substitute. Anything above 3.7 is not.

A set is 4 propellers: 2 clockwise + 2 counter-clockwise. Buying 2 sets gives one spare set.

Quotes — write the price here

Shop / unit no.	Exact item seen	Price S$	Bought?

Where to go

call before travelling

Components — Sim Lim Tower, 10 Jalan Besar

Shop	Known for	Buy here
J.C.W. Electronic	LED components, **DC fans**, power adapters	LED strip + fan — try here first, it may have both
Continental Electronic Enterprise	General hobby electronics	Resistors, wire, connectors
Kuriosity	Arduino / Raspberry Pi	Addressable LED strip, JST pigtails

Directory is ~2 years old — unit numbers may have changed. Most shops shut by ~7:30 pm regardless of posted hours. Prices vary between units; ask two or three.

Propellers — Sim Lim Square, 1 Rochor Canal Road

Hobby Square	FPV drone shop inside Sim Lim Square. Lists propellers from 1" to 12" including carbon fibre. Listed hours Mon–Sat 12:30 pm – 7:30 pm.

If Hobby Square has no 7037

Shop	Notes
Sky Hobbies skyhobbies.com.sg	Jalan Sultan area. Claims to be the largest HQProps stockist in Singapore.
Helicopter Smartfly smartfly.com.sg	Queensway Shopping Centre. Mon–Sat 12–7 pm, Sun 3–7 pm, **closed Wednesday**.
Rotor Hobby shop.rotor.com.sg	French Road, Kitchener Complex. Tel +65 6296 3686.

Fan fallback

Amicus amicus.com.sg	8 Ubi Road 2, #03-13 Zervex. Tel +65 6980 6783. Stocks 5 V brushless fans with order collection — **call first**, the listed model is 60 mm not 25 mm.